OpenADMET CYP Challenge — Time-Dependent Inhibition Track Model Description Report

Submitted by: Cheminfosolutions Ltd · **leaderboard alias:** Cheminfo
Predicted properties: `CYP3A4_is_TDI`, `CYP2D6_is_TDI`; blind set 750 compounds, training set 6,145 compounds
Metric: Matthews correlation coefficient (MCC)
Training data: challenge-distributed data only; no external TDI corpus was merged into any submitted column

1. Models

Models were built per isoform: CYP2D6 and CYP3A4 are independent runs with their own hyperparameters, blend weight and decision threshold. Selection was on MCC under Bemis–Murcko scaffold splits.

Arm	Model / architecture	Molecular representation	Reference	In final file
ChemProp	Multitask binary D-MPNN, Optuna-TPE tuned (depth, message/FFN widths, dropout, LR schedule, aggregation, auxiliary-head weight)	Learned bond-message graph embedding **+** RDKit 2D physicochemical descriptor block	Heid et al., *J. Chem. Inf. Model.* **2024**, 64, 9–17; Yang et al., *ibid.* **2019**, 59, 3370–3388	Yes
LightGBM	Gradient-boosted trees; tree count and decision threshold tuned on MCC	RDKit descriptors + Morgan (ECFP) counts + FCFP counts + MACCS keys + mechanism-based inactivation (MBI) alert counts	Ke et al., *NeurIPS* **2017**, 3146–3154	Yes (CYP3A4)
Published-protocol grid	kNN (Jaccard), Random Forest, SVM (RBF), ANN/MLP and a graph-convolution arm, crossed with 3 sampling strategies (original / ROS / RUS) and 10 fingerprints; 5-fold with per-fold tuning. Reproduced on the paper's 7,914-compound CYP3A4 set, then retrained on the challenge labels; the submitted column uses only the retrained form	PaDEL fingerprints: AP2D (780), CDK (1024), CDKExt (1024), EState (79), GraphFP (1024), MACCS (166), KRFP (4860), PubChemFP (881), SubFP (307)	Xu, Lu, Wu, Gui, Liu, Tang & Li, *Mol. Pharmaceutics* **2023**, 20, 194–205 (doi: 10.1021/acs.molpharmaceut.2c00571)	Yes
ChemProp, CheMeleon trunk	Same D-MPNN initialised from descriptor-pretrained foundation weights	as ChemProp	CheMeleon foundation weights distributed with ChemProp v2	No
XGBoost	Companion boosting arm on the same feature block as LightGBM	as LightGBM	Chen & Guestrin, *KDD* **2016**, 785–794	No
External-label auxiliary head	Multitask classifier with an auxiliary head on a published deep-learning TDI / experimental-variability set	as ChemProp	Fluetsch & Trunzer, *Chem. Res. Toxicol.* **2024**, 37, 549	No
Size controls	Heavy-atom count and molecular weight as single-feature rankers	—	internal controls	No

Descriptors. The in-house arms use the RDKit 2D physicochemical block (~200 descriptors) concatenated with count-based Morgan and FCFP fingerprints, MACCS keys and a hand-curated MBI structural-alert count vector; non-finite columns and columns exceeding 1E10 are dropped before fitting. The published-protocol grid uses PaDEL fingerprints exactly as published. Software: ChemProp, LightGBM, XGBoost, RDKit (Landrum, rdkit.org), PaDEL (Yap, *J. Comput. Chem.* **2011**, 32, 1466–1474), scikit-learn (Pedregosa et al., *JMLR* **2011**, 12, 2825–2830) and Optuna (Akiba et al., *KDD* **2019**, 2623–2631).

Submitted configuration.

Endpoint	Ranking	Filter	Positive calls / 750
CYP3A4	Rank average 0.6·rank(ChemProp) + 0.4·rank(LightGBM); weight read at its MCC maximum on a 0→1 sweep	Intersected with the published-protocol ranking, keeping compounds above the median of both; call count set at the MCC-optimal call rate implied by each endpoint's test AUC	194
CYP2D6	ChemProp ranking alone		54

2. Results

Endpoint	MCC	Accuracy	Precision	Recall
CYP2D6_is_TDI	**0.3895**	0.9153	0.4127	0.4625
CYP3A4_is_TDI	**0.4578**	0.7779	0.6227	0.6049
Macro-average	**0.4237**	—	—	—

Board read 2026-11-01; 750 rows, validated with the challenge's own submission validator.

3. Ligand preparation and labels

Ligand preparation, applied identically to the training and blind sets. The six RDKit corrections run in the order shown, through a single driver.

#	Step	Detail
0	Schrödinger LigPrep	Desalting, neutralisation, tautomer and ionisation enumeration at pH 7.4
1	Sulfonamide N	R-SO_2-[N^-]-R′ → R-SO_2-NH-R′
2	Pyridinium N	[nH^+] → n; quaternary [n^+] left unchanged
3	Morpholine N	[NH^+] → N
4	Iminium N	Amidine / guanidine / iminium C=[NH^+] → C=N, sp^2 nitrogen only
5	Amide	`N=C([O-])` → `NC(=O)`, bond orders corrected
6	Phenol	`c[O-]` → `cOH`
7	Artefact fallback	Unrepairable LigPrep artefacts revert to the raw input SMILES; raw, LigPrep and cleaned SMILES retained side by side, so the step is idempotent. Charged species 19.2% → 14.6%
8	Featuriser	Largest fragment selected, then RDKit `Uncharger`: graph and descriptor block are computed on neutral structures

Label interpretation. `is_TDI` is a deterministic function of two measured potency surfaces, verified to reproduce the published label on every training row carrying both.

Definition	`is_TDI` ⟺ `max(pIC50_TDI_condition, 4.0) − max(pIC50_direct, 4.0) > log10(2)`
Negative class	Any compound whose shifted curve never clears 4.301, so part of the class is compounds whose potency was never established rather than demonstrated non-inhibitors — 34.8% of the CYP3A4 negatives
Censoring	Left-censored potency rows treated as upper bounds, not as points
CYP2D6 coverage	No labelled compound in the saturated zone of the single-shot screen (curves were run only on pre-flagged compounds); that region filled with pseudo-negatives derived from the isoforms with data there, contamination 1 positive in 836 zone compounds